



UNITED INTERNATIONAL UNIVERSITY
 Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Microprocessors and Microcontrollers Laboratory
2	Course Code	CSE 4326
3	Trimester and Year	Summer 2026
4	Pre-requisites	EEE 2124 Electronics Laboratory
5	Credit Hours	1.0
6	Section	A
7	Class Hours	Saturday: 08:30 AM - 11:00 AM
8	Class Room	Digital Lab 1 (511)
9	Instructor's Name	Md. Shafqat Talukder Rakin (MSTR)
10	Email	shafqat@cse.uiu.ac.bd
11	Office	219 (A)
12	Counselling Hours	Updated in LMS (If you cannot locate me in my room during my counselling hour, just give me a call.)
13	Text Book	None.
14	Reference	None.
15	Course Contents (approved by UGC)	Students will design simple embedded systems using the principles learned in CSE 4325.

16	Course Outcomes (COs)	<table><tr><th>COs</th><th>Description</th><th>Bloom's Domain</th><th>Program Outcome</th><th>Knowledge Profile</th><th>Complex Engineering Problems</th><th>Complex Eng. Activities</th></tr><tr><td>CO 1</td><td>Illustrate the interfacing of microcontroller with different input/output devices and simulate using Proteus to observe and analyze behaviors of different electronic hardware circuits.</td><td>C</td><td>e <i>Modern Tool Usage</i></td><td rowspan="3">K3 Engineering Fundamentals K4 Specialist Knowledge K5 Design K6 Technology</td><td>(Any three) P1 Depth of Knowledge P2 Wide ranging P3 No obvious solution P7 Many Components</td><td>A1 Range of resources A2 Level of Interaction</td></tr><tr><td>CO 2</td><td>Design a basic embedded hardware group project.</td><td>A</td><td>c <i>Design of solutions</i></td></tr><tr><td>CO 3</td><td>Work in a team and communicate effectively.</td><td>A</td><td>i <i>Individual work and teamwork</i> j <i>Communication</i></td></tr></table>	COs	Description	Bloom's Domain	Program Outcome	Knowledge Profile	Complex Engineering Problems	Complex Eng. Activities	CO 1	Illustrate the interfacing of microcontroller with different input/output devices and simulate using Proteus to observe and analyze behaviors of different electronic hardware circuits.	C	e <i>Modern Tool Usage</i>	K3 Engineering Fundamentals K4 Specialist Knowledge K5 Design K6 Technology	(Any three) P1 Depth of Knowledge P2 Wide ranging P3 No obvious solution P7 Many Components	A1 Range of resources A2 Level of Interaction	CO 2	Design a basic embedded hardware group project.	A	c <i>Design of solutions</i>	CO 3	Work in a team and communicate effectively.	A	i <i>Individual work and teamwork</i> j <i>Communication</i>
COs	Description	Bloom's Domain	Program Outcome	Knowledge Profile	Complex Engineering Problems	Complex Eng. Activities																		
CO 1	Illustrate the interfacing of microcontroller with different input/output devices and simulate using Proteus to observe and analyze behaviors of different electronic hardware circuits.	C	e <i>Modern Tool Usage</i>	K3 Engineering Fundamentals K4 Specialist Knowledge K5 Design K6 Technology	(Any three) P1 Depth of Knowledge P2 Wide ranging P3 No obvious solution P7 Many Components	A1 Range of resources A2 Level of Interaction																		
CO 2	Design a basic embedded hardware group project.	A	c <i>Design of solutions</i>																					
CO 3	Work in a team and communicate effectively.	A	i <i>Individual work and teamwork</i> j <i>Communication</i>																					
17	Teaching Methods	Lecture, Laboratory exercise and reports, Project Developments.																						
18	CO with Assessment Methods	<table><tr><th>CO</th><th>Assessment Method</th><th>(%)</th></tr><tr><td>-</td><td>Attendance</td><td>5%</td></tr><tr><td>CO3</td><td>Class Performance</td><td>5%</td></tr><tr><td>CO3</td><td>Report/Viva</td><td>10%</td></tr><tr><td>CO2</td><td>Project</td><td>40%</td></tr><tr><td>CO1</td><td>Mid Quiz + Final Exam</td><td>20% + 20%</td></tr></table>	CO	Assessment Method	(%)	-	Attendance	5%	CO3	Class Performance	5%	CO3	Report/Viva	10%	CO2	Project	40%	CO1	Mid Quiz + Final Exam	20% + 20%				
CO	Assessment Method	(%)																						
-	Attendance	5%																						
CO3	Class Performance	5%																						
CO3	Report/Viva	10%																						
CO2	Project	40%																						
CO1	Mid Quiz + Final Exam	20% + 20%																						

20	Lab Outline																								
	<table><tr><th>Class</th><th>Topics/Assignments</th><th>COs</th><th>Lab Outcomes/Activities</th></tr><tr><td>Lab1</td><td>Experiment 1: An introduction to Arduino, Interfacing of Gas Sensor using Arduino & Showing the Sensor Data in OLED Display</td><td>CO1, CO2</td><td>To understand the configuration of Microcontroller Arduino board, learn how to code, connect arduino to perform simple projects and simulation of the such projects in Proteus.</td></tr><tr><td>Lab2</td><td>Experiment 1: An introduction to Arduino, Interfacing of Gas Sensor using Arduino & Showing the Sensor Data in OLED Display</td><td>CO1, CO2</td><td>To learn about gas sensor, understand how gas sensor works using Arduino. Also, learn about how OLED Display works. Interfacing Arduino, OLED Display and other sensors to perform different mini projects using Arduino. Also, learn how to use datasheet in real life experiment.</td></tr><tr><td>Lab3</td><td>Experiment 2: Wifi Communication and building IoT based systems using Arduino</td><td>CO1, CO2</td><td>To learn how to use microcontroller ESP32 using Arduino IoT Cloud. Also, how to interface gas sensor to send data via wifi to store and display on the cloud</td></tr><tr><td>Lab4</td><td>MID Quiz</td><td></td><td></td></tr><tr><td>Lab5</td><td>Experiment 3: Introduction to Raspberry Pi (Gen 4 Model B/B+)</td><td>CO1, CO2</td><td>To understand the configuration of the microprocessor Raspberry pi, how to</td></tr></table>	Class	Topics/Assignments	COs	Lab Outcomes/Activities	Lab1	Experiment 1: An introduction to Arduino, Interfacing of Gas Sensor using Arduino & Showing the Sensor Data in OLED Display	CO1, CO2	To understand the configuration of Microcontroller Arduino board, learn how to code, connect arduino to perform simple projects and simulation of the such projects in Proteus.	Lab2	Experiment 1: An introduction to Arduino, Interfacing of Gas Sensor using Arduino & Showing the Sensor Data in OLED Display	CO1, CO2	To learn about gas sensor, understand how gas sensor works using Arduino. Also, learn about how OLED Display works. Interfacing Arduino, OLED Display and other sensors to perform different mini projects using Arduino. Also, learn how to use datasheet in real life experiment.	Lab3	Experiment 2: Wifi Communication and building IoT based systems using Arduino	CO1, CO2	To learn how to use microcontroller ESP32 using Arduino IoT Cloud. Also, how to interface gas sensor to send data via wifi to store and display on the cloud	Lab4	MID Quiz			Lab5	Experiment 3: Introduction to Raspberry Pi (Gen 4 Model B/B+)	CO1, CO2	To understand the configuration of the microprocessor Raspberry pi, how to
Class	Topics/Assignments	COs	Lab Outcomes/Activities																						
Lab1	Experiment 1: An introduction to Arduino, Interfacing of Gas Sensor using Arduino & Showing the Sensor Data in OLED Display	CO1, CO2	To understand the configuration of Microcontroller Arduino board, learn how to code, connect arduino to perform simple projects and simulation of the such projects in Proteus.																						
Lab2	Experiment 1: An introduction to Arduino, Interfacing of Gas Sensor using Arduino & Showing the Sensor Data in OLED Display	CO1, CO2	To learn about gas sensor, understand how gas sensor works using Arduino. Also, learn about how OLED Display works. Interfacing Arduino, OLED Display and other sensors to perform different mini projects using Arduino. Also, learn how to use datasheet in real life experiment.																						
Lab3	Experiment 2: Wifi Communication and building IoT based systems using Arduino	CO1, CO2	To learn how to use microcontroller ESP32 using Arduino IoT Cloud. Also, how to interface gas sensor to send data via wifi to store and display on the cloud																						
Lab4	MID Quiz																								
Lab5	Experiment 3: Introduction to Raspberry Pi (Gen 4 Model B/B+)	CO1, CO2	To understand the configuration of the microprocessor Raspberry pi, how to																						

					interface with Led using the GPIO of Pi.	
		Lab6	Experiment No 4: Video processing using Raspberry Pi	CO1, CO2	To understand the configuration of Pi camera and how to use it using Raspberry Pi. Also, learn how simple computer vision problems can be solved using pi.	
		Lab7	Final Exam			
		Lab8	Final Project Proposal	CO3		
		Lab9	Project Update 1	CO3		
		Lab10	Project Update 2	CO3		
		Lab11	Optional Updates	CO3		
		Lab12	Final Presentation and Project showing	CO3		

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	5%
Class Performance	5%
Report/Viva	10%
Project	40%
Mid Quiz + Final Exam	20% + 20%

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

	Program Outcomes
1	Engineering knowledge: Apply knowledge of mathematics, natural science, engineering fundamentals and Computer Science and Engineering to the solution of complex engineering problems.
2	Problem analysis: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3	Design/development of solutions: Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4	Investigation: Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions
5	Modern tool usage: Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.
6	The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.
7	Environment and sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.
8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
9	Individual work and teamwork: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	Project management and finance: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.
----	--